Tangential Convergence of the Central Path in Nonlinear Semidefinite Optimization Without Nondegeneracy

A verification and repair note

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Abstract

Okuno asked whether the derivative of the primal central path converges to its first-order displacement vector under the strict-complementarity, enhanced second-order, and Mangasarian–Fromovitz assumptions used in his analysis of nonlinear semidefinite optimization:

$$\lim_{\mu \downarrow 0} \dot{x}(\mu) = \xi^*.$$

We prove that it does. The argument does not differentiate the known quotient limit $(x(\mu) - x^*)/\mu \to \xi^*$. Under the stated C^2 assumptions that quotient need not possess a finite one-sided derivative, as an explicit example shows. Instead, after eliminating equality constraints, we factor the log-determinant barrier through a fixed-face Schur complement. Differentiation along the central path gives an exact singularly perturbed linear system. Its normal block is controlled by the active-face log-determinant Hessian, while its tangential block is controlled by the enhanced second-order condition, including the semidefinite sigma term. A finite-dimensional perturbation lemma then yields convergence of the derivative without requiring a rate for the dual convergence. Subject to the hypotheses and the central-path and quotient-limit results proved in Okuno's paper, this is a complete solution of the primal tangential-derivative conjecture.

Status. The result below is a *complete solution*, rather than partial progress, within Okuno's standing framework. It also repairs the central defect of an earlier attempted proof: the rescaled quotient path generally is not C^1 at $\mu = 0$.

1 Setting and inherited results

Consider the nonlinear semidefinite program

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad G(x) \in \mathbb{S}^m_+, \qquad h(x) = 0, \tag{1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$, $G : \mathbb{R}^n \to \mathbb{S}^m$, and $h : \mathbb{R}^n \to \mathbb{R}^s$ are twice continuously differentiable. The Lagrangian is

$$L(x, Y, z) = f(x) - G(x) \bullet Y + h(x)^T z, \qquad X \bullet Y := \operatorname{tr}(XY).$$

For $d \in \mathbb{R}^n$, write $DG(x)[d]$ for the directional derivative of G, and let $DG(x)^*$ be its adjoint for the Euclidean and Frobenius inner products.

Let x^* be the KKT point considered in [1]. We impose Okuno's Assumption 1:

(A1) strict complementarity holds at x^* for at least one multiplier;

(A2) the enhanced second-order sufficient condition holds for every multiplier in the multiplier set at x^*;

(A3) the Mangasarian–Fromovitz constraint qualification holds at x^*.

The enhanced condition states that, for each (Y, z) in the multiplier set,

$$d^T(\nabla^2_{xx} L(x^*, Y, z) + \Omega(x^*, Y))d > 0 \qquad (d \in \mathcal{C}(x^*) \setminus \{0\}), \tag{2}$$

where $\mathcal{C}(x^*)$ is the critical cone and Ω is the semidefinite sigma-term matrix.

Okuno proves that, for some $\bar{\mu} > 0$, there is a unique smooth central path

$$w(\mu) = (x(\mu), Y(\mu), z(\mu)), \qquad 0 < \mu < \bar{\mu},$$

satisfying

$$\nabla_x L(x(\mu), Y(\mu), z(\mu)) = 0, \tag{3}$$

$$G(x(\mu))Y(\mu) = \mu I, \tag{4}$$

$$h(x(\mu)) = 0, \tag{5}$$

with $G(x(\mu)), Y(\mu) \succ 0$, and

$$w(\mu) \longrightarrow w_a := (x^*, Y_a, z_a). \tag{6}$$

Here (Y_a, z_a) is the analytic-center multiplier selected by the barrier.

Choose an orthogonal matrix $P = [E\ F]$ such that

$$P^T G(x^*) P = \begin{bmatrix} 0 & 0 \\ 0 & C_* \end{bmatrix}, \qquad P^T Y_a P = \begin{bmatrix} Y_0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad C_* \succ 0, \quad Y_0 \succ 0. \tag{7}$$

The E-block has size $k := m - \operatorname{rank} G(x^*)$. Empty blocks and their determinants are interpreted in the standard way.

Okuno defines ξ^* as the unique Δx-component of the limiting system

$$U_*^T(\nabla^2_{xx} L(w_a)\Delta x - DG(x^*)^* \Delta Y) = 0, \tag{8}$$

$$\mathcal{L}_{G(x^*)}(\Delta Y) + \mathcal{L}_{Y_a}(DG(x^*)[\Delta x]) = 2I, \tag{9}$$

$$\nabla h(x^*)^T \Delta x = 0, \tag{10}$$

where $\mathcal{L}_X(Z) = XZ + ZX$, and the columns of U_* form an orthonormal basis of

$$\mathcal{U}_* := \{d : \nabla h(x^*)^T d = 0,\ E^T DG(x^*)[d]E = 0\}. \tag{11}$$

The active block of (9) gives

$$E^T DG(x^*)[\xi^*]E = Y_0^{-1}. \tag{12}$$

The quotient theorem in [1] gives

$$\frac{x(\mu) - x^*}{\mu} \longrightarrow \xi^*. \tag{13}$$

The open question is whether the derivative has the same limit.

Theorem 1.1 (Tangential convergence)**.** *Under assumptions* (A1)–(A3)*,*

$$\lim_{\mu \downarrow 0} \dot{x}(\mu) = \xi^*. \tag{14}$$

2 Equality reduction and the fixed-face Schur complement

By MFCQ, $\nabla h(x^*)$ has full column rank. The constant-rank theorem therefore gives a neighborhood $V \subset \mathbb{R}^q$, where $q = n - s$, a point $v^* \in V$, and a C^2 parametrization

$$\Phi : V \to \mathbb{R}^n, \qquad \Phi(v^*) = x^*, \qquad h(\Phi(v)) = 0,$$

which is a diffeomorphism onto a neighborhood of x^* in the equality manifold. Put

$$J := D\Phi(v^*) \in \mathbb{R}^{n\times q}, \qquad \widehat{f} := f \circ \Phi, \qquad \widehat{G}(v) := P^T G(\Phi(v))P.$$

The range of J is $\{d : \nabla h(x^*)^T d = 0\}$.

Write

$$\widehat{G}(v) = \begin{bmatrix} A(v) & B(v) \\ B(v)^T & C(v) \end{bmatrix}. \tag{15}$$

At v^*, $A(v^*) = 0$, $B(v^*) = 0$, and $C(v^*) = C_* \succ 0$. Shrinking V if necessary, $C(v) \succ 0$ throughout V. Define the fixed-face Schur complement

$$\Sigma(v) := A(v) - B(v)C(v)^{-1}B(v)^T. \tag{16}$$

Whenever $\widehat{G}(v) \succ 0$,

$$\det \widehat{G}(v) = \det C(v) \det \Sigma(v), \qquad \Sigma(v) \succ 0. \tag{17}$$

Let $v_\mu := \Phi^{-1}(x(\mu))$. From (13), there is a unique $p^* \in \mathbb{R}^q$ such that

$$v_\mu = v^* + \mu p^* + o(\mu), \qquad Jp^* = \xi^*. \tag{18}$$

Define

$$\mathscr{A} := D\Sigma(v^*) : \mathbb{R}^q \to \mathbb{S}^k, \qquad \mathcal{U} := \ker \mathscr{A}, \qquad \mathcal{N} := \mathcal{U}^\perp. \tag{19}$$

Since $B(v^*) = 0$,

$$\mathscr{A}p = E^T DG(x^*)[Jp]E. \tag{20}$$

Thus (12) gives

$$\mathscr{A}p^* = M_* := Y_0^{-1}. \tag{21}$$

Finally, define

$$H_* := \nabla^2\big(\widehat{f} - \langle Y_0, \Sigma\rangle\big)(v^*), \qquad c_* := \nabla \log\det C(v^*). \tag{22}$$

If the F-block is empty, $\log\det C$ is understood to be zero.

2.1 The Schur complement contains the sigma term

For $d, e \in \mathbb{R}^n$, define the bilinear polarization of the sigma term at (x^*, Y_a) by

$$\begin{aligned} \omega_a(d,e) := {} & \operatorname{tr}\Big(Y_0\, DG^{EF}(x^*)[d]C_*^{-1}DG^{FE}(x^*)[e]\Big) \\ & + \operatorname{tr}\Big(Y_0\, DG^{EF}(x^*)[e]C_*^{-1}DG^{FE}(x^*)[d]\Big), \end{aligned} \tag{23}$$

where $DG^{EF}[d] = E^T DG[d]F$ and $DG^{FE}[d] = (DG^{EF}[d])^T$. Okuno's block formula for the sigma term is

$$\omega_a(d,d) = d^T\Omega(x^*, Y_a)d. \tag{24}$$

Lemma 2.1 (Schur-complement Hessian)**.** *For $p, q \in \mathbb{R}^q$, with $d = Jp$ and $e = Jq$,*

$$p^T H_* q = d^T \nabla^2_{xx} L(w_a) e + \omega_a(d, e). \tag{25}$$

Consequently, there exists $\beta > 0$ such that

$$u^T H_* u \geq \beta \, \|u\|^2 \qquad (u \in \mathcal{U}). \tag{26}$$

Proof. Because $h \circ \Phi = 0$ and $P^T Y_a P = \mathrm{diag}(Y_0, 0)$,

$$\widehat{f}(v) - \langle Y_0, A(v) \rangle = L(\Phi(v), Y_a, z_a).$$

Since $\nabla_x L(w_a) = 0$, differentiating twice at v^* gives, for $d = Jp$ and $e = Jq$,

$$D^2(\widehat{f} - \langle Y_0, A \rangle)(v^*)[p, q] = d^T \nabla^2_{xx} L(w_a) e. \tag{27}$$

At v^* we have $B = 0$, and hence

$$\begin{aligned} D^2(BC^{-1}B^T)(v^*)[p, q] = DB(v^*)[p] C_*^{-1} DB(v^*)[q]^T \\ + DB(v^*)[q] C_*^{-1} DB(v^*)[p]^T. \end{aligned}$$

Combining this identity with $\Sigma = A - BC^{-1}B^T$ and (27) proves (25).

Now let $0 \neq u \in \mathcal{U}$ and put $d = Ju$. Then $\nabla h(x^*)^T d = 0$ and $E^T DG(x^*)[d] E = 0$. Thus $DG(x^*)[d]$ belongs to the tangent cone of $\mathbb{S}^m_+$ at $G(x^*)$. KKT stationarity also gives

$$\nabla f(x^*)^T d = Y_a \bullet DG(x^*)[d] - z_a^T \nabla h(x^*)^T d = 0.$$

Hence $d \in \mathcal{C}(x^*)$. Since J has full column rank, $d \neq 0$, and (2) at (Y_a, z_a) yields $u^T H_* u > 0$. Compactness of the unit sphere in $\mathcal{U}$ gives (26). ☐

2.2 A reduced characterization of the first-order displacement

Let $P_{\mathcal{U}}$ and $P_{\mathcal{N}}$ denote the orthogonal projections associated with (19).

Lemma 2.2 (Characterization of p^*)**.** *The vector p^* is the unique solution of*

$$\mathscr{A} p = M_*, \tag{28}$$

$$P_{\mathcal{U}}(H_* p - c_*) = 0. \tag{29}$$

Proof. The normal equation is (21). We derive the tangential equation from reduced barrier stationarity, without differentiating the quotient path.

For $\mu > 0$, set

$$M_\mu := \frac{\Sigma(v_\mu)}{\mu} \in \mathbb{S}^k_{++}, \qquad \mathscr{A}_\mu := D\Sigma(v_\mu), \qquad c_\mu := \nabla \log\det C(v_\mu). \tag{30}$$

The top-left block of $\widehat{G}(v_\mu)^{-1}$ is $\Sigma(v_\mu)^{-1}$. Since $P^T Y(\mu) P = \mu \widehat{G}(v_\mu)^{-1}$, we have the exact identity

$$Y(\mu)^{EE} = M_\mu^{-1}. \tag{31}$$

Thus (6) implies

$$M_\mu^{-1} \to Y_0, \qquad M_\mu \to M_* = Y_0^{-1}. \tag{32}$$

Using (17), reduced barrier stationarity becomes the regular equation

$$\nabla \widehat{f}(v_\mu) - \mathscr{A}_\mu^* M_\mu^{-1} - \mu c_\mu = 0. \tag{33}$$

Reduced KKT stationarity at v^* is

$$\nabla \widehat{f}(v^*) - \mathscr{A}^* Y_0 = 0. \tag{34}$$

Because $P_{\mathcal{U}} \mathscr{A}^* = 0$, subtracting (34) from (33), projecting onto $\mathcal{U}$, and dividing by μ gives

$$0 = P_{\mathcal{U}} \left[\frac{\nabla \widehat{f}(v_\mu) - \nabla \widehat{f}(v^*)}{\mu} - \frac{(\mathscr{A}_\mu - \mathscr{A})^* M_\mu^{-1}}{\mu} - c_\mu \right]. \tag{35}$$

Since Σ is C^2 and (18) holds,

$$\mathscr{A}_\mu = \mathscr{A} + \mu \mathscr{B} + o(\mu), \qquad \mathscr{B}[q] := D^2\Sigma(v^*)[p^*, q]. \tag{36}$$

Passing to the limit in (35) yields

$$P_{\mathcal{U}}(\nabla^2 \widehat{f}(v^*) p^* - \mathscr{B}^* Y_0 - c_*) = 0. \tag{37}$$

For every $u \in \mathbb{R}^q$,

$$u^T \mathscr{B}^* Y_0 = \left\langle Y_0, D^2\Sigma(v^*)[p^*, u] \right\rangle = u^T \nabla^2 \langle Y_0, \Sigma \rangle (v^*) p^*.$$

Hence (37) is exactly (29).

For uniqueness, suppose p_1, p_2 satisfy (28)–(29). Then $\delta = p_1 - p_2 \in \mathcal{U}$ and $P_{\mathcal{U}} H_* \delta = 0$. Thus $\delta^T H_* \delta = 0$, and Lemma 2.1 implies $\delta = 0$. $\square$

3 A finite-dimensional singular-perturbation lemma

The next lemma isolates the normal–tangential mechanism. It assumes only convergence of Q_ε, y_ε, and r_ε; no first-order rates for these quantities are needed.

Lemma 3.1 (Normal–tangential singular perturbation)**.** *Let $\mathcal{X}$ and $\mathcal{Z}$ be finite-dimensional real Hilbert spaces. Let*

$$A : \mathcal{X} \to \mathcal{Z}, \qquad \mathcal{U} := \ker A, \qquad \mathcal{N} := \mathcal{U}^\perp,$$

and let $P_{\mathcal{U}}, P_{\mathcal{N}}$ be the corresponding orthogonal projections. Suppose:

(i) $Q : \mathcal{Z} \to \mathcal{Z}$ is self-adjoint and positive definite;

(ii) $H : \mathcal{X} \to \mathcal{X}$ is self-adjoint and positive definite on $\mathcal{U}$;

(iii) $m \in \operatorname{range} A$, and $y = Qm$;

(iv) as $\varepsilon \downarrow 0$,

$$A_\varepsilon = A + \varepsilon B + o(\varepsilon), \quad Q_\varepsilon \to Q, \quad H_\varepsilon \to H, \quad y_\varepsilon \to y, \quad r_\varepsilon \to r,$$

where each Q_ε is self-adjoint positive definite and each H_ε is self-adjoint.

If $d_\varepsilon \in \mathcal{X}$ satisfies

$$(A_\varepsilon^* Q_\varepsilon A_\varepsilon + \varepsilon H_\varepsilon) d_\varepsilon = A_\varepsilon^* y_\varepsilon + \varepsilon r_\varepsilon, \tag{38}$$

then d_ε converges to the unique vector d_* satisfying

$$A d_* = m, \qquad P_{\mathcal{U}}(H d_* - r) = 0. \tag{39}$$

Proof. Write $d_\varepsilon = n_\varepsilon + u_\varepsilon$ with $n_\varepsilon \in \mathcal{N}$ and $u_\varepsilon \in \mathcal{U}$, and put $K_\varepsilon = A_\varepsilon^* Q_\varepsilon A_\varepsilon$. Since $A|_{\mathcal{N}}$ is injective and Q is positive definite, $P_{\mathcal{N}} A^* Q A P_{\mathcal{N}}$ is positive definite on $\mathcal{N}$. Also $A_\varepsilon P_{\mathcal{U}} = O(\varepsilon)$, so

$$P_{\mathcal{N}} K_\varepsilon P_{\mathcal{N}} \longrightarrow P_{\mathcal{N}} A^* Q A P_{\mathcal{N}}, \tag{40}$$

$$P_{\mathcal{N}} K_\varepsilon P_{\mathcal{U}} = O(\varepsilon), \tag{41}$$

$$P_{\mathcal{U}} K_\varepsilon P_{\mathcal{U}} = O(\varepsilon^2), \tag{42}$$

$$P_{\mathcal{U}} A_\varepsilon^* y_\varepsilon = O(\varepsilon). \tag{43}$$

Furthermore, $P_{\mathcal{U}} H_\varepsilon P_{\mathcal{U}}$ is uniformly positive definite on $\mathcal{U}$ for small ε.

The $\mathcal{N}$-block of (38) gives

$$\|n_\varepsilon\| \le C(1 + \varepsilon \|u_\varepsilon\|). \tag{44}$$

After dividing the $\mathcal{U}$-block by ε, the coefficient of u_ε is

$$\varepsilon^{-1} P_{\mathcal{U}} K_\varepsilon P_{\mathcal{U}} + P_{\mathcal{U}} H_\varepsilon P_{\mathcal{U}}. \tag{45}$$

The first term is positive semidefinite and the second is uniformly positive definite. All remaining coefficients and the divided right-hand side are bounded by (41)–(43). Hence

$$\|u_\varepsilon\| \le C(1 + \|n_\varepsilon\|). \tag{46}$$

Combining (44) and (46), and absorbing the resulting $O(\varepsilon)\,\|n_\varepsilon\|$ term, proves boundedness of $\{d_\varepsilon\}$.

Take a convergent subsequence, still indexed by ε, with $n_\varepsilon \to n$ and $u_\varepsilon \to u$. The limit of the $\mathcal{N}$-block is

$$A^* Q A n = A^* y = A^* Q m.$$

Choose $n_0 \in \mathcal{N}$ with $A n_0 = m$, possible because $m \in \operatorname{range} A$. Taking the inner product with $n - n_0$ gives

$$\langle An - m, Q(An - m)\rangle = 0,$$

so $An = m$.

For the $\mathcal{U}$-block divided by ε, note that

$$\frac{P_{\mathcal{U}} A_\varepsilon^*}{\varepsilon} = P_{\mathcal{U}} \left(\frac{A_\varepsilon - A}{\varepsilon} \right)^* \longrightarrow P_{\mathcal{U}} B^*.$$

Also, $A_\varepsilon u_\varepsilon = O(\varepsilon)$ and $P_{\mathcal{U}} A_\varepsilon^* = O(\varepsilon)$. Since $An = m$ and $y = Qm$,

$$\begin{aligned} \varepsilon^{-1} P_{\mathcal{U}} K_\varepsilon n_\varepsilon &\longrightarrow P_{\mathcal{U}} B^* Q A n = P_{\mathcal{U}} B^* y, \\ \varepsilon^{-1} P_{\mathcal{U}} K_\varepsilon u_\varepsilon &\longrightarrow 0, \\ \varepsilon^{-1} P_{\mathcal{U}} A_\varepsilon^* y_\varepsilon &\longrightarrow P_{\mathcal{U}} B^* y. \end{aligned}$$

The two $B^* y$ terms cancel, leaving

$$P_{\mathcal{U}}(H(n + u) - r) = 0.$$

Thus every accumulation point satisfies (39).

If d_1, d_2 are two solutions of (39), then $\delta = d_1 - d_2 \in \mathcal{U}$ and $P_{\mathcal{U}} H \delta = 0$. Therefore $\langle \delta, H\delta \rangle = 0$, and positivity of H on $\mathcal{U}$ gives $\delta = 0$. The accumulation point is unique, so the full family converges. □

4 Exact derivative equation and proof of the conjecture

Proof of Theorem 1.1. Assume first that $k > 0$. Retain the notation

$$M_\mu = \frac{\Sigma(v_\mu)}{\mu}, \qquad \mathscr{A}_\mu = D\Sigma(v_\mu), \qquad c_\mu = \nabla \log\det C(v_\mu).$$

By (32), $M_\mu \to M_* = Y_0^{-1}$. Define

$$\mathcal{Q}_\mu[Z] := M_\mu^{-1} Z M_\mu^{-1}, \qquad Z \in \mathbb{S}^k. \tag{47}$$

This operator is self-adjoint and positive definite for the Frobenius inner product, because

$$Z \bullet \mathcal{Q}_\mu[Z] = \left\| M_\mu^{-1/2} Z M_\mu^{-1/2} \right\|_F^2.$$

Define the self-adjoint operator

$$H_\mu := \nabla^2 \Big(\widehat{f} - \Big\langle M_\mu^{-1}, \Sigma \Big\rangle - \mu \log\det C \Big) (v_\mu), \tag{48}$$

where M_μ^{-1} is held fixed while taking the Hessian. Differentiating (33) along the path gives

$$H_\mu \dot{v}_\mu + \mathscr{A}_\mu^* \mathcal{Q}_\mu[\dot{M}_\mu] = c_\mu. \tag{49}$$

Indeed, $D(M^{-1})[\dot{M}] = -M^{-1}\dot{M}M^{-1}$, and the minus sign in stationarity changes this to the plus sign in (49).

Differentiating $M_\mu = \Sigma(v_\mu)/\mu$ yields

$$\mu \dot{M}_\mu = \mathscr{A}_\mu \dot{v}_\mu - M_\mu. \tag{50}$$

Substitution into (49) gives the exact equation

$$(\mathscr{A}_\mu^* \mathcal{Q}_\mu \mathscr{A}_\mu + \mu H_\mu)\dot{v}_\mu = \mathscr{A}_\mu^* M_\mu^{-1} + \mu c_\mu. \tag{51}$$

The required limits are

$$\mathscr{A}_\mu = \mathscr{A} + \mu \mathscr{B} + o(\mu), \qquad \mathscr{B}[q] = D^2\Sigma(v^*)[p^*, q], \tag{52}$$

$$\mathcal{Q}_\mu \to \mathcal{Q}_* := [Z \mapsto Y_0 Z Y_0], \qquad H_\mu \to H_*, \qquad c_\mu \to c_*, \tag{53}$$

$$M_\mu^{-1} \to Y_0 = \mathcal{Q}_*[M_*]. \tag{54}$$

The first line follows from C^2 regularity and (18); the remaining limits follow from (32) and continuity of the second derivatives. Also $M_* \in \operatorname{range} \mathscr{A}$ by (21), and H_* is positive definite on $\ker \mathscr{A}$ by Lemma 2.1.

Apply Lemma 3.1 to (51) with

$$\mathcal{X} = \mathbb{R}^q, \quad \mathcal{Z} = \mathbb{S}^k, \quad A = \mathscr{A}, \quad Q = \mathcal{Q}_*, \quad H = H_*, \quad m = M_*, \quad y = Y_0, \quad r = c_*.$$

It follows that $\dot{v}_\mu$ converges to the unique vector p satisfying

$$\mathscr{A} p = M_*, \qquad P_{\mathcal{U}}(H_* p - c_*) = 0.$$

By Lemma 2.2, this vector is p^*. Therefore

$$\dot{v}_\mu \longrightarrow p^*. \tag{55}$$

Since $x(\mu) = \Phi(v_\mu)$,

$$\dot{x}(\mu) = D\Phi(v_\mu)\dot{v}_\mu \longrightarrow D\Phi(v^*)p^* = \xi^*.$$

If $k = 0$, then $G(x^*) \succ 0$ and $Y_a = 0$. Reduced stationarity is

$$\nabla \widehat{f}(v_\mu) - \mu \nabla \log\det \widehat{G}(v_\mu) = 0.$$

The enhanced second-order condition makes $\nabla^2 \widehat{f}(v^*)$ positive definite on the whole reduced space. Differentiation gives

$$\left(\nabla^2 \widehat{f}(v_\mu) - \mu \nabla^2 \log\det \widehat{G}(v_\mu)\right) \dot{v}_\mu = \nabla \log\det \widehat{G}(v_\mu),$$

so

$$\dot{v}_\mu \longrightarrow \nabla^2 \widehat{f}(v^*)^{-1} \nabla \log\det \widehat{G}(v^*).$$

Subtracting stationarity at v^*, dividing by μ, and using (18) identifies the same vector with p^*. Hence $\dot{x}(\mu) \to \xi^*$ also in this case. □

Corollary 4.1. *Let $p(\mu) = (v_\mu - v^*)/\mu$. Then*

$$\mu\dot{p}(\mu) \longrightarrow 0.$$

No boundedness of $\dot{p}(\mu)$ is asserted or needed.

Proof. Since $\dot{v}_\mu = p(\mu) + \mu\dot{p}(\mu)$, combine $p(\mu) \to p^*$ with $\dot{v}_\mu \to p^*$. □

Remark 4.2 (Regularity used)**.** *No third derivatives occur. The proof differentiates the C^1 reduced stationarity equation only for $\mu > 0$, uses continuity of the Hessians, and uses the first-order expansion of $D\Sigma$ supplied by $\Sigma \in C^2$. Thus Okuno's original C^2 data assumptions suffice.*

5 Independent checks and failure of the quotient-IFT approach

5.1 The quotient path need not be differentiable

Example 5.1. *Consider the scalar semidefinite problem*

$$\min_{x_1,x_2} \quad f(x_1,x_2) = x_1 + x_1^2 - x_1x_2 + \frac{1}{2}x_2^2 + \frac{2}{5}|x_2|^{5/2} \quad \textit{subject to} \quad x_1 \geq 0. \tag{56}$$

The objective is C^2, and

$$f = x_1 + \frac{1}{2}x_1^2 + \frac{1}{2}(x_1 - x_2)^2 + \frac{2}{5}|x_2|^{5/2},$$

so $x^* = (0,0)$ is the unique minimizer. The multiplier is $Y_a = 1$, strict complementarity and MFCQ hold, and the critical subspace is $\{d : d_1 = 0\}$. Since

$$\nabla^2 f(0,0) = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix},$$

the enhanced second-order condition holds on that subspace.

Parameterize the central path by $t = x_2 > 0$. The BKKT equations give

$$x_1 = t + t^{3/2}, \qquad Y = 1 + t + 2t^{3/2}, \qquad \mu = (t + t^{3/2})(1 + t + 2t^{3/2}). \tag{57}$$

Thus $\xi^* = (1,1)$, whereas the second quotient component is

$$p_2(\mu) = \frac{t}{\mu} = \frac{1}{(1+\sqrt{t})(1+t+2t^{3/2})} = 1 - \sqrt{\mu} + O(\mu). \tag{58}$$

Consequently $\dot{p}_2(\mu) \sim -1/(2\sqrt{\mu})$, so the quotient path has no finite one-sided derivative at zero. Nevertheless, differentiating (57) with respect to t gives

$$\frac{\mathrm{d}\mu}{\mathrm{d}t} \to 1, \qquad \frac{\mathrm{d}x_2}{\mathrm{d}\mu} \to 1, \qquad \frac{\mathrm{d}x_1}{\mathrm{d}\mu} \to 1.$$

Thus $\dot{x}(\mu) \to (1,1) = \xi^*$, as Theorem 1.1 predicts.

5.2 An exact check with failure of nondegeneracy

Example 5.2. *Fix $\alpha \in \mathbb{R}$ and consider*

$$\min_{x,y} \quad x + \alpha xy + \frac{1}{2}y^2 \quad \text{subject to} \quad G(x,y) = \operatorname{diag}(x,x) \succeq 0. \tag{59}$$

At $(0,0)$, MFCQ and strict complementarity hold, but nondegeneracy fails because the two active diagonal constraints have identical gradients. The multiplier set is

$$\{\operatorname{diag}(\lambda_1,\lambda_2) : \lambda_1,\lambda_2 \geq 0,\ \lambda_1+\lambda_2 = 1\},$$

and its analytic center is $Y_a = \frac{1}{2}I$. Critical directions have $d_x = 0$, and the enhanced second-order form is d_y^2, so the standing assumptions hold locally.

The reduced barrier is

$$x + \alpha xy + \frac{1}{2}y^2 - 2\mu \log x.$$

Its stationarity equations are

$$y = -\alpha x, \qquad x - \alpha^2 x^2 = 2\mu.$$

For the branch converging to the origin,

$$x(\mu) = 2\mu + O(\mu^2), \qquad y(\mu) = -2\alpha\mu + O(\mu^2),$$

and direct differentiation gives

$$\dot{x}(\mu) \to 2, \qquad \dot{y}(\mu) \to -2\alpha.$$

Here $\mathscr{A}(d_x,d_y) = \operatorname{diag}(d_x,d_x)$ has non-surjective range, $M_* = Y_0^{-1} = 2I$, and the normal–tangential equations give $p^* = (2,-2\alpha)$. This checks that the proof does not impose nondegeneracy or surjectivity of $\mathscr{A}$.

5.3 The sigma term is indispensable

Example 5.3. *Consider*

$$\min_{x_1,x_2} \quad x_1 - \frac{1}{2}x_2^2 \quad \text{subject to} \quad \begin{bmatrix} x_1 & x_2 \\ x_2 & 1 \end{bmatrix} \succeq 0. \tag{60}$$

The feasible set is $x_1 \geq x_2^2$, so $(0,0)$ is a strict minimizer. At the analytic-center multiplier $Y_a = \operatorname{diag}(1,0)$, the Lagrangian Hessian contributes $-d_2^2$ on the critical direction $d = (0,d_2)$, while the sigma term contributes $2d_2^2$. Their sum is $d_2^2 > 0$.

The Schur complement is $\Sigma(x) = x_1 - x_2^2$, and barrier stationarity gives

$$x_1(\mu) = \mu, \qquad x_2(\mu) = 0.$$

Hence $\dot{x}(\mu) = (1,0) = \xi^*$. This verifies both the sign and the factor in Lemma 2.1. Any coercivity argument using only the Lagrangian Hessian would fail on this example.

6 Conclusion

The conjectured limit is valid under Okuno's original framework. The decisive point is not differentiability of the quotient path at the boundary. The fixed-face Schur complement instead converts barrier stationarity into a regular equation involving the scaled active Schur complement M_μ. Its derivative has two scales: an order-one normal block and an order-μ tangential block. The active log-determinant Hessian controls the former; the enhanced second-order condition, with the full semidefinite sigma term, controls the latter. The cancellation in Lemma 3.1 requires only $M_\mu \to Y_0^{-1}$, equivalently $Y(\mu)^{EE} \to Y_0$, and not a first-order rate.

Accordingly, the note supplies a complete solution of the primal tangential-derivative problem. It does not assert convergence of all components of $\dot{Y}(\mu)$, nor a C^1 extension of the rescaled full primal–dual path; those are separate questions.

References

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